

Enough water to matter

A briefing · 9 August 2026 · Vancouver, BC

Figures are generated from the simulation at pinkrobotics.ca/airships and verified against it automatically.

THE IDEA

Water is how fires are stopped. The only questions have ever been how much of it you can put on one, how fast, and for how long — and the answers today are *not enough, not fast enough, and not for long*. This is a design for a machine that changes all three at once.

The problem is the round trip

A very large airtanker carries about seventy tonnes. It arrives, releases in seconds, and then spends the next half hour to hour-and-a-half flying to a base, loading, and flying back. Its instantaneous delivery is spectacular; its *sustained* delivery — the number that decides whether a fire line holds — is set almost entirely by the turnaround.

Helicopters with buckets shorten the turnaround by dipping from a lake nearby, and pay for it in payload: a heavy-lift helicopter carries about ten tonnes.

Nothing occupies the gap between them — a machine that dips like a helicopter and carries like a tanker — because for a rotorcraft every kilogram of lift is bought continuously with power, so payload and endurance trade directly against each other.

Buoyant flight does not have that trade. A hull that displaces its own weight in air holds altitude at zero power. Carrying more costs the price of a bigger hull, once. Put a buoyant vehicle over a lake and the turnaround stops existing: it never lands, so there is nothing to turn around. Nobody is aboard — these are uncrewed ships, sized for the British Columbia interior, where the fires and the lakes are usually within tens of kilometres of each other.

What the model says a fleet would do

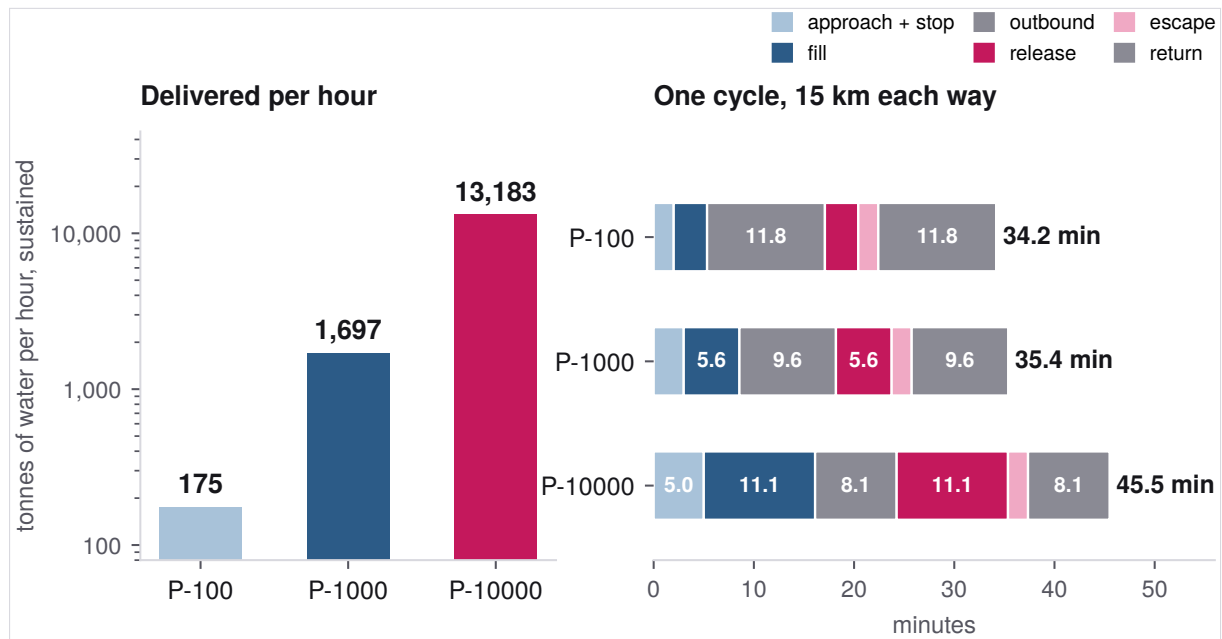


Figure 1. Sustained delivery, and the cycle that produces it. Nothing here lands: on every class the longest phases are moving water, not flying.

	P-100	P-1000	P-10000
Water per drop	100 t	1,000 t	10,000 t
Length	190 m	404 m	876 m
Round trip, 15 km each way	34.2 min	35.4 min	45.5 min
Water delivered per hour	175 t	1,697 t	13,183 t
Energy per tonne delivered	12.5 kWh	7.4 kWh	4.6 kWh

Thirteen thousand tonnes an hour is thirteen million litres an hour, from one aircraft, without landing, through the night. **Nothing in that table has been built** — no hull, no rotor, no cable. It is what the model says a fleet would do, which is why the model is published and why the last third of this page is the four things that would have to be true. At industrial electricity prices, the energy in a full 10,000 -tonne drop costs on the order of two thousand dollars. **It also gets cheaper as it gets bigger:** buoyancy scales with volume and drag with area, so the largest class delivers a tonne for 4.6 kWh against the smallest class's 12.5. The square-cube law, which punishes almost every other kind of vehicle, is on this one's side.

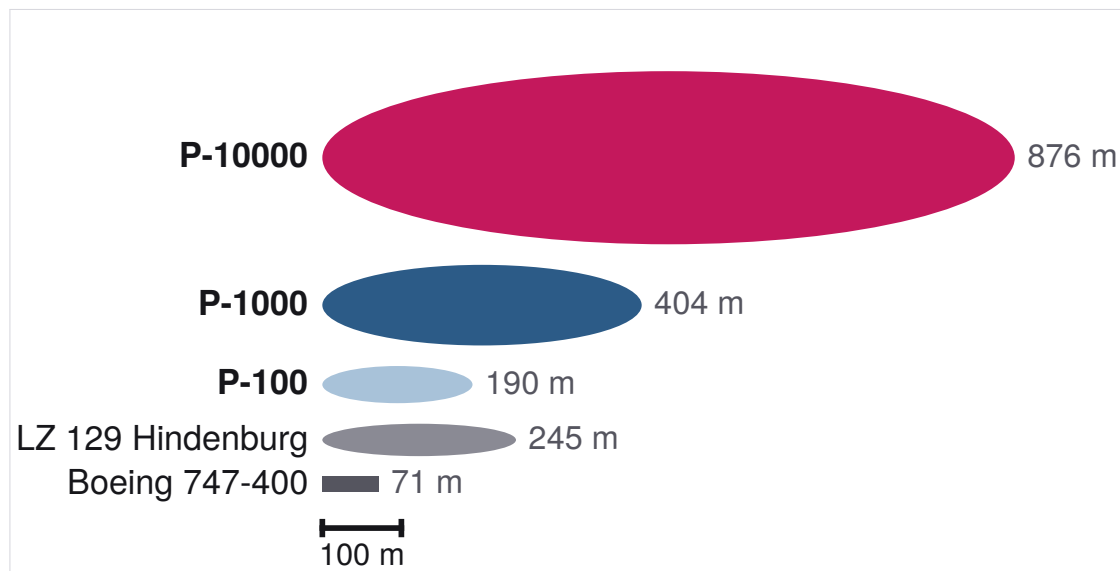


Figure 2. The three classes at true relative scale, against the two largest aircraft most readers can picture. The volume is the lift.

Why it is possible now, and was not before

A conventional airship floats because it is full of helium. This one floats because it is full of *nothing* — a rigid shell holding a vacuum, about 14% more buoyant than helium, and impossible to embargo, to corner, or to leak.

The idea is 356 years old. Francesco Lana de Terzi published it in 1670 and was right about everything except the metal: thin the shell enough to float and the atmosphere crushes it. That was never fixable by waiting for better materials. Akhmeteli and Gavrilin put a number on it in 2021 — floating and surviving together demand a stiffness-to-density ratio **no solid substance possesses**, diamond included.

What changed is not a better material but a better *arrangement* of ordinary ones. Discrete lattice structures — ultralight frameworks assembled from many identical mass-produced parts — hold their stiffness as they lose density, where foams and honeycombs do not. Jenett, Gregg and Cheung, at NASA Ames and MIT, applied that to the vacuum balloon and found the binding constraint moves from **buckling to strength**. Buckling is a geometry failure: sudden, total, unfixable by a better material. Strength is a number you can look up, test on a bench, and buy more of. The question stopped being *is this impossible* and became *how light can we build it* — and that is an engineering question, which is the kind that gets answered.

Borrowing the lake

One part of this is not a scaling exercise, and it is our favourite thing in the project. A hull big enough to float when it is full of water is very hard to push *down* when it is empty — hardest of all at the bottom, over the lake, where the air is thickest. Our largest class has to hold down 13,723 t of surplus lift there and its rotors can manage 12,666. It cannot reach its own water under power. The answer is to borrow the lake. The ship lowers a cable with a collapsible bag, fills it at the surface, and winches it just clear: **12,400 tonnes hanging on a line is 12,400 tonnes of downward force**, and the whole price of it is the fifteen metres of lift needed to break the surface. When the ship has what it came for, the bag is tipped back into the lake it came from.

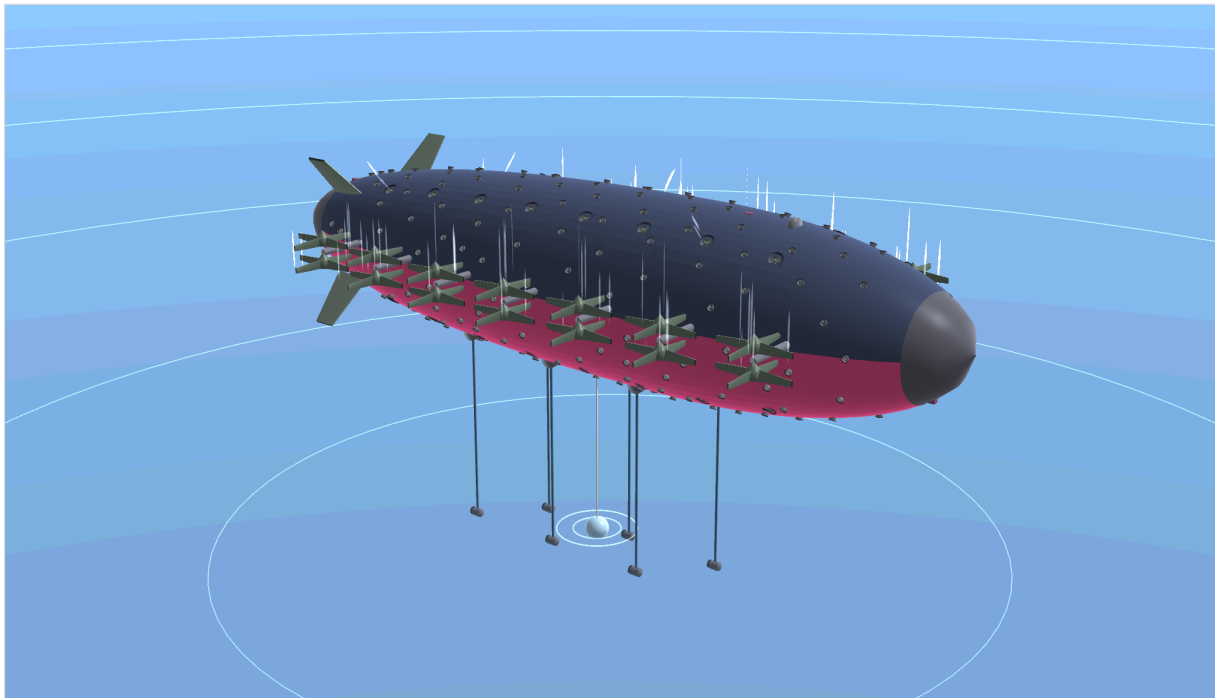


Figure 3. The mechanism, rendered from the same model that computes the numbers: pump pods on hoses to the surface, and the anchor cable running down to a bag sitting in the water. The bag is to scale — 12,400 tonnes is 28.7 m across beside an 876 m hull. The rotors are pushing the ship *down* here, which is why the wash blows up.

It costs 0.596 MWh and removes 34.5 MWh of rotor work. **Fifty-eight to one.** Rotors get disproportionately expensive as you load them, so every tonne handed to the lake is worth more than a tonne taken off the rotors. That is why the bag does the *whole* descent rather than just covering the shortfall that revealed it, and why every class carries one — including the class whose rotors could manage without. The alternatives were costed in the open and rejected: the same ballast as liquid nitrogen is 475 MWh, and filling from high up a 1,350 m hose is 44 MWh at two-metre bore and 140 bar. When one mechanism is that much cheaper than every alternative, that is usually the design telling you something.

What would have to be true

This is arithmetic on a vehicle nobody has built, resting on assumptions we have gone looking for evidence *against*. Four are load-bearing:

- **A structure light enough.** The model assumes a ship's dry mass equals its water payload. NASA's own lattice paper — the one this rests on — gives a bare shell 12% heavier than that whole allowance, and at the battery density NASA has flown, the battery alone is 34% over. **This is what the concept lives or dies on.** It looks like a manufacturing question rather than a physics one — which is the good news — and it is unanswered.
- **A drop that arrives.** The US Forest Service says a release 1,000 ft above the canopy "would completely dissipate". Ours release from 1,476 ft, raised for hull clearance and never costed against delivery.
- **Ground crews.** The Forest Service measures whether fire behaviour changed, not tonnes, and finds success turns on crews being engaged with the drop. Tonnage is our metric; it may be the wrong one.
- **An energy chain.** Every class runs a deficit every cycle — a P-100 has 10.4 hours of work in it, a P-10000 36.3. This fleet is a battery being spent.

Seventeen such issues are tracked at `docs/OPEN-QUESTIONS.md`, fifteen still open. Two closed on 2026-08-09, and both closed by making our own numbers *worse*.

Why we publish the arithmetic

Because a concept nobody can check is not a concept, it is a picture. Every number here is produced by code you can run; every claim cites a figure the model generates, and the build fails if the prose and the model disagree. Two tests fail on purpose, so a known defect cannot quietly lose its excuse. We catalogue the sources that contradict us beside the ones that support us — nine against, at last count.

One finding from that discipline is worth more than any single number. When we corrected the solar skin, the vehicle's power supply changed by a factor of 4.4 and **not one published figure moved**, because generation was not in the ledger those figures come from. A model can be badly wrong about something central with nothing able to notice. We would rather be corrected early and loudly than be right in private.

WHERE THIS GOES

If a shell this light can be built, what falls out of it is not only a firefighting machine. Moving ten thousand tonnes of water for four and a half kilowatt-hours a tonne is a capability nothing currently has, at any price. Fire is where it would start: the need is loudest there, and the round trip is shortest.

Live model pinkrobotics.ca/airships

The evidence pinkrobotics.ca/research

Repository, defect list and 73 catalogued sources github.com/pinkrobotics/airships

Figures here are generated from the model at defaults — balanced mode, 15 km one way — and checked against it on every build.